Data Visualization - Matplotlib

Data Visualization

We, as humans, process visual information faster than text or numbers, so we use **Data visualization** i.e. the graphical representation of data to:

- Understand patterns and trends
- Identify outliers and anomalies
- Communicate insights clearly
- Support data-driven decision making

The common types of plots used in data visualization are line plots, bar charts, histograms, scatter plots, box plots etc.

Matplotlib

Matplotlib is a **Python library** used for:

- Creating static, animated, and interactive plots
- Low-level control over plot appearance
- Serving as the foundation for other libraries (Seaborn, Pandas plotting)

Installation

```
pip install matplotlib
```

Usage

```
import matplotlib.pyplot as plt
```

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Basic Plot Structure

```
plt.plot(x, y)
plt.xlabel("X-axis label")
plt.ylabel("Y-axis label")
plt.title("Title")
plt.show()
```

Important Plots in Matplotlib

Line Plots

A line plot is a type of chart used to display data points connected by straight lines, typically to show trends or changes over a continuous variable, such as time or distance.

Example:

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

plt.plot(x, y)
plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Simple Line Plot")
plt.grid(True) # Adds a grid
plt.show()
```

Key features of a line plot:

- **X-axis**: usually continuous (time, sequence, measurements)
- **Y-axis**: values being measured
- **Line style**: solid (), dashed (`-`), dotted (`:`) etc.
- **Markers**: shows individual data points (`o` , `s` , `^` , `'*'` , `'.'`)

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Styling a Line Plot

```
plt.plot(x, y,
         color="blue",
         linestyle="--",
         linewidth=2,
         marker="o",
         markersize=6)

plt.show()
```

Multiple Lines on the Same Plot

```
y2 = [1, 3, 5, 7, 9]

plt.plot(x, y, label="Line 1")
plt.plot(x, y2, label="Line 2")

plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()
```

Saving Plots

```
plt.savefig("line_plot.png")
```

When to use?

Use a line plot when:

- Data points are ordered
- You want to show progression or trends
- You're comparing changes rather than individual categories

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In Matplotlib, `fmt` strings are a compact way to specify the line style, marker, and color in a single string, so `fmt` is a combination of `[color][marker][linestyle]`.

Example:

```
plt.plot(x, y, 'g--') # green dashed line
plt.plot(x, y, 'bo')  # blue circles, no line
```

Refer to documentation for all options - https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.plot.html

Bar Charts

A bar chart is a graph that displays categorical data using rectangular bars. Each bar's height (or length, in horizontal bars) represents a value.

Example:

```
import matplotlib.pyplot as plt

x = ['A', 'B', 'C', 'D']
y = [10, 15, 7, 12]

plt.bar(x, y)
plt.xlabel("Categories")
plt.ylabel("Values")
plt.title("Simple Bar Chart")
plt.show()
```

This creates a vertical bar chart (most common).

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Styling a Bar Chart

```
plt.bar(x, y,
        color='skyblue',
        width=0.6,
        edgecolor='black',
        linewidth=1.5)

plt.text()
plt.show()
```

Horizontal Bar Chart

```
plt.barh(x, y)
plt.xlabel("Values")
plt.ylabel("Categories")
plt.title("Horizontal Bar Chart")
plt.show()
```

Multiple Bar Charts (Grouped Bars)

```
import numpy as np

x = np.arange(4)
y1 = [10, 15, 7, 12]
y2 = [8, 14, 9, 10]

width = 0.35

plt.bar(x - width/2, y1, width, label='Group 1')
plt.bar(x + width/2, y2, width, label='Group 2')

plt.xticks(x, ['A', 'B', 'C', 'D'])
plt.xlabel("Categories")
plt.ylabel("Values")
plt.legend()
plt.show()
```

When to use?

Use a bar chart when:

- You are comparing **different categories**
- The data is **discrete**, not continuous
- You want clear visual comparison

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We can use `plt.text()` to add text annotations to the bars in the bar chart.

Syntax:

```
values = [10, 20, 30]
plt.bar(['A', 'B', 'C'], values)

for i, v in enumerate(values):
plt.text(i, v, str(v))
```

Scatter Plots

A scatter plot shows the relationship between two variables by plotting points on a 2D plane.

Example:

```
x = [1, 2, 3, 4, 5]
y = [2, 4, 1, 3, 5]

plt.scatter(x, y)
plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Simple Scatter Plot")
plt.grid(True)
plt.show()
```

Styling a Bar Chart

```
plt.scatter(x, y,
            s=100,            # marker size
            c='red',          # color
            marker='o',
            alpha= 0.7)       # transparency
plt.show()
```

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Using a Colormap (Color by Value)

```
import numpy as np

colors = np.array([10, 20, 30, 40, 50])

plt.scatter(x, y, c=colors, cmap='viridis')
plt.colorbar(label='Color Scale')
plt.show()
```

Multiple Scatter Plots

```
y2 = [5, 3, 4, 2, 1]

plt.scatter(x, y, label='Set 1')
plt.scatter(x, y2, label='Set 2')

plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()
```

When to use?

Use a scatter plot when:

- You want to analyze the **relationship** between two variables
- Data is **numerical and continuous**
- You don't want to imply a trend by connecting points



We can also add annotations. **Annotations** are used to add explanatory text or markers to specific points in a plot.

Example:

```
x = [1, 2, 3, 4, 5]
y = [5, 7, 6, 8, 7]

plt.scatter(x, y)

# Annotate each point
for i in range(len(x)):
    plt.text(x[i]+0.1, y[i]+0.1, f"({x[i]}, {y[i]})") # small offset
```

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Pie Charts

A pie chart is a circular chart divided into slices to show relative proportions of a whole.

- Each slice = a category
- Size of slice = value of that category relative to the total

We prefer them only when our data has **few categories**.

Example:

```
sizes = [30, 20, 25, 25]
labels = ['A', 'B', 'C', 'D']

plt.pie(sizes, labels=labels)
plt.show()
```

Adding Percentages

```
plt.pie(sizes,
        labels=labels,
        autopct='%1.1f%%') # shows percentage

plt.show()
```

Changing Colors

```
colors = ['skyblue', 'lightgreen', 'pink', 'orange']

plt.pie(sizes, labels=labels, autopct='%1.1f%%', colors=colors)
plt.show()
```

Starting Angle and Shadow

```
plt.pie(sizes,
        labels=labels,
        autopct='%1.1f%%',
        startangle=90,  # rotate chart
        shadow=True)    # adds shadow

plt.show()
```

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Wedgeprops

In Matplotlib pie charts, `wedgeprops` is a dictionary of properties that controls the appearance of the pie slices (wedges). It has common properties like edgecolor, linewidth, linestyle, alpha etc.

```
plt.pie(sizes,
        labels=labels,
        autopct='%1.1f%%',
        wedgeprops={'edgecolor': 'black', 'linewidth': 2})
```

When to use?

Use a pie chart when:

- You want to display **part-to-whole relationships**
- You have **categorical data**
- Data has **few categories** (4–6); too many slices make it hard to read

Histograms

A histogram shows the distribution of numerical data by:

- Dividing values into **bins** (ranges)
- Counting how many values fall into each bin (frequency)

Example:

```
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5]

plt.hist(data, bins=5, color='skyblue', edgecolor='black')

plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Simple Histogram")
plt.grid(axis='y', linestyle='--', alpha=0. 7)

plt.show()
```

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Number of bins

`bins` controls how the data is divided. It can be an integer or a sequence of bin edges:

```
plt.hist(data, bins=3) # integer

plt.hist(data, bins=[1, 2, 3, 4, 5, 6]) # sequence
```

Histogram Orientation

- Vertical (default)

```
plt.hist(data)
```

- Horizontal

```
plt.hist(data, orientation='horizontal')
```

Multiple Histograms

```
data2 = [2, 3, 3, 4, 4, 5, 5, 5, 6]

plt.hist(data, bins=5, alpha=0.5, label='Data 1', color='blue')
plt.hist(data2, bins=5, alpha=0.5, label='Data 2', color='red')
plt.legend()
```

`alpha` controls transparency so overlapping histograms are visible.

When to use?

Use a histogram when:

- You want to show the **distribution of continuous data**.
- You have to analyze **frequency, spread, skewness**, or patterns.
- Working with **large datasets** to summarize trends.

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`axvline` (Axis Vertical Line) is used to draw a **vertical line** at a specific x-value in a plot.

Example:

```
plt.axvline(x=value, color='color', linestyle='style', linewidth=width, label='label')
```

Box Plots

A box plot is a statistical visualization that summarizes the distribution of a dataset using five key numbers:

- Minimum (lowest non-outlier)
- First Quartile (Q1) – 25th percentile
- Median (Q2) – 50th percentile
- Third Quartile (Q3) – 75th percentile
- Maximum (highest non-outlier)

It can also show **outliers**.

Example:

```
data = [7, 8, 5, 6, 9, 7, 8, 10, 4, 6]

plt.boxplot(data)
plt.ylabel("Values")
plt.title("Simple Box Plot")
plt.show()
```

Main Values in the Box Plot

1. **Minimum (Lower Whisker)**
   - The smallest data point within 1.5 × IQR below Q1
   - Any smaller points are considered outliers

2. **First Quartile (Q1)**
   - The 25th percentile of the data
   - 25% of data points are below Q1
   - Bottom edge of the box

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3. **Median (Q2)**
   - The 50th percentile (middle value)
   - Line inside the box
   - Splits the dataset into two halves

4. **Third Quartile (Q3)**
   - The 75th percentile of the data
   - 75% of data points are below Q3
   - Top edge of the box

5. **Maximum (Upper Whisker)**
   - The largest data point within 1.5 × IQR above Q3
   - Points beyond this are outliers

6. **Interquartile Range (IQR)**
   - Difference between Q3 and Q1: $\text{IQR} = \text{Q3} - \text{Q1}$
   - Represents the middle 50% of the data

7. **Outliers**
   - Data points outside 1.5 × IQR from Q1 or Q3
   - Plotted as dots or asterisks beyond the whiskers

Horizontal Box Plot

```
plt.boxplot(data, vert=False)
plt.xlabel("Values")
plt.show()
```

Multiple Box Plots

```
data1 = [7, 8, 5, 6, 9, 7, 8, 10, 4, 6]
data2 = [5, 6, 7, 8, 5, 4, 6, 7, 5, 6]

plt.boxplot([data1, data2], labels=['Dataset 1', 'Dataset 2'])
plt.show()
```

Each box represents a different dataset.

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Showing Mean

```
plt.boxplot(data, showmeans=True, meanline=True)
```

- `showmeans=True` adds the mean marker.
- `meanline=True` draws a line instead of a point.

When to use?

- Visualize spread and skewness of data
- Identify outliers
- Compare distributions across multiple groups

Stack Plots

A stack plot is a type of plot where multiple data series are stacked on top of each other. Each “layer” shows the contribution of one category, and the top line shows the cumulative total.

Example:

```
x = [1, 2, 3, 4, 5]
y1 = [1, 2, 3, 4, 5]
y2 = [2, 1, 2, 1, 2]

plt.stackplot(x, y1, y2, labels=['Data1', 'Data2'], colors=['blue', 'green'])

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Basic Stack Plot")
plt.legend(loc='upper left')

plt.show()
```

When to use?

Use a stack plot when you want to:

- Show cumulative data over time
- Compare multiple components
- Highlight trends in parts and whole

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Subplots

In Matplotlib, subplots allow you to display multiple plots in a single figure, arranged in a grid. This is useful for comparing different datasets or visualizations side by side.

Example:

```
x = [1, 2, 3, 4, 5]
y1 = [2, 3, 5, 7, 11]
y2 = [1, 4, 2, 5, 3]

# 1 row, 2 columns, first subplot
plt.subplot(1, 2, 1) # (rows, columns, index)
plt.plot(x, y1, color='blue', marker='o')
plt.title("First Subplot")
plt.xlabel ("X")
plt.ylabel("Y1")

# 1 row, 2 columns, second subplot
plt.subplot(1, 2, 2)
plt.plot(x, y2, color='red', marker='s')
plt.title("Second Subplot")
plt.xlabel("X")
plt.ylabel("Y2")

plt.tight_layout()
plt.show()
```

- `plt.subplot(nrows, ncols, index)` selects the current axes.
- Index counts row-wise from top-left to bottom-right (Row major).

Modern Matplotlib

Modern Matplotlib with the **Object-Oriented (OO)** approach is now the recommended way to create professional plots.

Why Use OO Style?

- More control over multiple axes, subplots, and figures
- Cleaner and more readable for complex plots
- Avoids side effects of `plt` (state-based interface)
- Easier to combine multiple plot types

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Basic Plot

```
x = [1, 2, 3, 4, 5]
y = [2, 3, 5, 7, 11]

# Create figure and axes
fig, ax= plt.subplots()

# Plot data
ax.plot(x, y, label="Prime numbers", color='blue', linestyle='--', marker='o')

# Add labels and title
ax.set_xlabel("X axis")
ax.set_ylabel("Y axis")
ax.set_title("OO Line Plot Example")

# Add legend and grid
ax.legend()
ax.grid(True)

plt.show()
```

`fig` and `ax` are core objects that give you full control over our plots.

- `fig` is Figure & it represents the entire figure or canvas.
- `ax` is Axes & it represents a single plot or graph within the figure.

Subplots

```
fig, axs = plt.subplots(2, 2, figsize=(8, 6))

axs[0, 0].plot(x, y1)
axs[0, 0].set_title("Top Left")

axs[0, 1].bar(['A','B','C'], [3,5,2])
axs[0, 1].set_title("Top Right")

axs[1, 0].scatter(x, y2)
axs[1, 0].set_title("Bottom Left")

axs[1, 1].hist([1,2,2,3,3,3,4])
axs[1, 1].set_title("Bottom Right")

plt.tight_layout()
plt.show()
```

> *Keep Learning & Keep Exploring!*

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